

Zakk Lefkowitz

SENIOR SOFTWARE ENGINEER

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Summary

Senior Software Engineer at **Amazon Fire TV**. 9 years in distributed systems, mobile-cloud architecture, and data engineering. Launched the Discovery Catalog data platform; previously shipped Continue Watching to **100M+ devices** as client-side tech lead. Consistent record of championing the harder, better architectural decision to build robust systems at Amazon scale. Pushing agentic engineering practice at Fire TV; built and operate a self-iterating agent orchestrator shipping **~3x** prior-year output of peer-reviewed code.

Experience

Amazon, Fire TV

Richmond, VA

SENIOR SOFTWARE ENGINEER

April 2020 - Present

Tech lead for Catalog data platforms and enrichment, previously client-side tech lead on Partner Integrations. Promoted to Sr. Software Engineer (Q4 2022).

Discovery Catalog Platform

Distributed Systems, Data Engineering, Java, S3, DynamoDB, RocksDB

- Designed and built a governed data platform extending the Fire TV Catalog beyond streaming-provider feeds to enriched sources (Wikipedia articles, scene descriptions), replacing fragmented ad-hoc pulls by downstream teams.
- Drove a separate publication architecture to protect the ~150 GB customer-critical core catalog from projected **terabyte-scale** ancillary data and enable a forward-compatible schema contract.
- Built fragment-based storage with raw S3 payloads hydrated on demand for curation, linking, and publication.
- Delivered to production solo in three months against a documented 4.5-month estimate. Enabled by agentic engineering.

Catalog Metadata Distillation & LLM Enrichment

LLM Applications & Evaluation, AWS Bedrock, Data Engineering

- Built metadata distillation that attaches canonical values to content clusters; launched with LLM-curated genres.
- Built an LLM genre classifier within Fire TV's ~100M-entity stateful incremental catalog pipeline at **~\$0.001 per entity**; filtered inference to the ~5% of updates most likely to reach customer screens, protecting end-to-end latency.
- Invented **CMAR (Catalog Metadata Acceptance Rate)**: an LLM-judge evaluation applied to 10K catalog entities after achieving >90% agreement with PM ratings on a 200-item calibration set. Classifier hit **95% CMAR vs. 82% baseline**; the metric earned PM confidence and unlocked downstream customer-team adoption.
- Tech lead on subsequent enrichment features shipped by mid-level engineers; core contributor to the org's shared Bedrock integration library.

Third-Party Continue Watching on Fire TV

Mobile-Cloud Architecture, Rust, Android, Java Services

- Led the client-side architecture on the VP-level initiative that gave Fire TV visibility into what customers watched inside partner apps, enabling Continue Watching and **8-figure incremental ad revenue**.
- Drove the decision to build the on-device middleware in Rust rather than piggyback on existing Android services. Shipped to 100M+ devices at <5MB memory footprint, proving out the native-service pattern subsequently adopted across Fire TV.
- Drove the decision to build a custom Amazon SDK rather than adopt Android's standard Watch Next API. Owned the entire partner-facing surface end-to-end (SDK, [integration docs](#), [reference sample app](#)); the resulting API surface became a partner-cited deal-closing factor across **Disney+, HBO Max, Hulu, Starz, and MGM+**.
- Built and operated the cloud ingestion backend handling sensitive customer and partner data at **~200K TPS peak**.
- Owned the privacy/consent surface end-to-end; copy experimentation drove **opt-in from 30% → 75%** in months.

Agentic Engineering

Agentic Workflow Automation, MCP

- Authored and maintain **catalog-engineer**: a shared agent spec (context + MCP servers + skills) adopted 100% across a 3-team, ~40-person org, cited by VP as a best-practice example, and replicated by sister Fire TV orgs.
- Built and operate an internal **human-gated autonomous engineering pipeline**. Shipping production code through Amazon's standard peer code review. **Q1 2026 alone shipped ~70% as many merged PRs as all of 2025** (~3x output multiplier). Shared the system in an org-wide talk to 100+ engineers, driving 10+ direct adopters and two VP follow-ups.
- Trusted by Fire TV VP leadership to help provision a new AI Engineering velocity team; subsequently drove Catalog org adoption of a code package context graph system, landing **100 core packages** in 30 days.

CapTech Consulting

Richmond, VA

SENIOR CONSULTANT

July 2017 - March 2020

Led cloud modernization initiatives for Global Top 10 insurance client. Promoted to Senior Consultant (Q3 2019).

Insurance Cloud Modernization

AWS ECS, CodeDeploy, Terraform, TypeScript, Angular, Node.js

- Architected **AWS ECS** migration and **CodeDeploy**-based CI/CD for a legacy Windows-based Java payment application processing ~\$1bn/year; built supporting **Terraform** IaC and a TypeScript deployment CLI.

Skills

Languages & Technologies
AWS, Cloud & Infrastructure
Architecture & Core
Competencies

Rust, Java, Kotlin, TypeScript, Python, SQL, Android, Spring Boot, Apache Spark, RocksDB, MCP
EC2, ECS, Lambda, DynamoDB, SQS, Bedrock, CDK, S3, CloudWatch, CodeDeploy, Terraform (IaC)
Distributed Systems, Data Engineering, Mobile-Cloud Architecture, Agentic Workflow Automation,
LLM Applications & Evaluation

Education

Virginia Polytechnic Institute and State University

Blacksburg, VA

B.S. COMPUTER SCIENCE, MATHEMATICS MINOR, CUM LAUDE

May 2017